



Soroush Haghi

Computer Engineering | Software & AI | Quantum Technologies

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PROFESSIONAL PROFILE

M.Sc. student in Quantum Technologies at TU Braunschweig with a B.Sc. in Computer Engineering (AI specialization). Practical Python, MATLAB, and software-engineering work spans machine learning, computer vision, sensor and signal processing, embedded projects, CI/CD, and technical automation. Academic work covers communication and coding systems, photonics, semiconductor/GaN technology, and quantum information; experience includes industry-university engineering, IT support, and supervised teaching.

EDUCATION

M.Sc. Quantum Technologies in Electrical and Computer Engineering

Oct 2024 - Present

Technische Universität Braunschweig, Germany | In progress (Internship and Thesis)

Selected coursework: Quantum Communication Networks; Information & Coding Theory with Computational Experiments; Network Information Theory; Nonlinear Photonics; Semiconductor/GaN Technology; Quantum Information & Computing

Academic engagement: presentations on QKD satellite communication, state-dependent channel coding, Quantum Communication Networks, and semiconductor crystal defects; plasmonic-metasurface journal-club literature review

B.Sc. Computer Engineering (Professionalization subject: AI)

2018 - Feb 2024

Azad University, Mashhad Branch, Iran

Overall grade: 17.28/20 | Academic standing: Ranked 5th out of 131 students (Top 4% of cohort)

Relevant focus: Artificial Intelligence; Embedded Systems; C/C++ programming; SQL and database coursework

SELECTED ACADEMIC TECHNICAL WORK

Coding Theory Computational Group Lab

MATLAB-based group work

- Analyzed cyclic-code structure using Hamming distance and correction radius, and examined recorded bit-error behavior versus SNR and related error-correction concepts.

Nonlinear Photonics Analytical Coursework

- Completed analytical work on dispersion, phase matching, harmonic generation, nonlinear optical effects, and fiber nonlinearities, including refractive-index and gain/threshold calculations.

Information & Coding Theory Graded Coursework

- Completed graded work on binary/block coding, prefix codes and Kraft sums, relative entropy, Markov processes, and typical sets.

TECHNICAL EXPERIENCE

Teaching assistant for Theory of Formal Languages and Automata

Sep 2023 - Jan 2024

Islamic Azad University, Mashhad Branch

- Led weekly exercise and problem-solving sessions for approximately 20 students.
- Corrected and evaluated assignments and exams within the supervised course.

Industry-University Project Member - Pouyesh Scheme

Apr 2023 - Sep 2023

Iran Khodro context | University Co-op

- Participated through the university in a MATLAB computer-vision vehicle-detection project.
- Ranked 1st among 50 participants in the relevant Pouyesh context.
- Subsequently received an invitation to work from Iran Khodro.

IT Support Intern

Jul 2022 - Sep 2022

Islamic Azad University, Mashhad, Iran

- Troubleshoot and repaired desktop computer hardware.
- Operating-system and system setup for staff and students.
- Provided staff and student support for local-network.

TECHNICAL SKILLS

Programming	Python; C; C++; MATLAB
Machine Learning & Computer Vision	PyTorch; scikit-learn; OpenCV; YOLO
Scientific Computing & Signal Processing	NumPy; SciPy; pandas; Matplotlib
Embedded Systems	ATmega32; Raspberry Pi
Development Tools & Databases	Git; GitHub/GitLab; Docker; CI/CD; pre-commit; SQL

SELECTED SOFTWARE & ENGINEERING PROJECTS

Career OS - AI-Assisted Career Workflow Automation ([Link](#))

Google Apps Script | Google Drive API | Gemini API | GitHub

- Built a privacy-first workflow for career-memory, evidence tracking, source ingestion, and application management, with change scanning, content fingerprints, duplicate-suppressed queues, OCR/audio transcription, manifests, and retry handling.

Gas Classification Pipeline and Simulation Dashboard ([Link](#))

Python | pandas | NumPy | scikit-learn | RandomForest | Streamlit

- Constructed statistical features from multi-sensor time windows, used a stratified train/test workflow for six-gas classification with probability output, and integrated the pipeline into a Streamlit dashboard.

Activity Recognition and CI Configuration ([Link](#))

Python | PyTorch | scikit-learn | GitLab CI | pytest | flake8 | Poetry

- Compared RandomForest and 1D-CNN approaches for sensor-activity recognition and configured dependency management, linting, tests, and GitLab CI quality checks.

MRI Segmentation Inference Dashboard ([Link](#))

PyTorch | Streamlit

- Integrated pretrained U-Net++ and EfficientNet-B4 inference with preprocessing, weight loading, execution, and visualization of masks, overlays, and outputs.

Day/Night Vehicle Detection ([Link](#))

MATLAB | Computer Vision | Image Processing | Bachelor Engineering Projects

- Built a road-intersection video-processing pipeline with day/night scene routing, Gaussian-mixture foreground detection, morphological processing, connected-component analysis, and region-based vehicle filtering.

Hybrid Image Generation ([Link](#)) | Stereo Sound Direction ([Link](#))

MATLAB | Computer Vision | Image and Signal Processing

- Combined low-frequency content from one image with high-frequency detail from another; separately processed stereo audio in one-second windows and used left/right RMS energy to classify the dominant sound direction.

ATmega32 Digital Clock ([Link](#)) | ATmega32 Digital Safe ([Link](#))

Embedded C | ATmega32 | 7-segment display | matrix keypad | LCD

- Implemented a multiplexed digital clock with rollover and forward/reverse time adjustment, plus a digital safe with keypad scanning, password comparison, LCD feedback, and locked/open state logic.

HONORS

Iran National University Entrance Exam (Konkoor) 2018: ~10,000 of ~140,000 (Top 7%)

Khorasan Razavi selective-school Entrance Exam 2014: 12 of ~27,000 (Top 1%)

LANGUAGES & CERTIFICATES

Languages	German - A2 course completed; exam result pending English – C1 TOEFL iBT 95/120 Persian - Native
Certificates	Quantum Machine Learning (QML) - Arya Quanta Institute Machine Learning; Introduction to Programming Using Python - Kaggle Jan 2025 Machine Learning and Data Science Bootcamp - Islamic Azad University, Mashhad Branch 96 hours Sep 2023 Python Programming - Jahad Daneshgahi, Khorasan Razavi Branch 40 hours 2021